

HAZOP ACTION TRACKING REGISTER

HAZOP-CDU2-2024 | Node 7 | Column Bottoms Circuit - Action Tracking Register

Study reference	HAZOP-CDU2-2024	Node	Node 7
Node description	C-2101 bottoms nozzle to E-2107 outlet, including P-2104A/B	Revalidation due	2029-06-30
Register extracted	2026-09-03	Facilitator	N. Bhat

Status summary: 2 Closed, 1 In Progress, 2 Not Started, **1 OVERDUE**. Overdue actions are reportable in the monthly process safety review.

Action	Deviation	Cause	Consequence	Existing safeguard	Recommendation	Owner	Due	Status
H7-01	More Flow	Control valve FCV-2141 fails open	Overload of downstream E-2107, tube vibration	Flow high alarm FAH-2141 at DCS	Confirm FCV-2141 fail-safe position on next overhaul	V. Nayak	2025-11-30	Closed
H7-02	No Flow	Both P-2104A and P-2104B unavailable	Column bottoms level high, potential carryover to C-2101 flash zone	Level high-high trip LSHH-2101 trips F-2101	Verify LSHH-2101 trip test frequency is 6 monthly, not annual	P. D'Souza	2026-03-31	Closed
H7-03	High Pressure	Blocked discharge with pump running	Overpressure of discharge header above design 24 kg/cm2g	PSV-2141 on discharge header	Re-rate PSV-2141: relief load recalculation shows required capacity exceeds installed capacity by an estimated 8%	V. Nayak	2026-06-30	OVERDUE
H7-04	Loss of Containment	Mechanical seal failure in hot RCO service	Hydrocarbon release at grade, potential auto-ignition	Area gas detectors GD-2104-1/2, fire water monitor coverage	Review seal selection against API 682 for services above 300 degC	S. Pai	2026-09-30	In Progress
H7-05	High Temperature	E-2107 bypass valve passing	RCO to storage above 150 degC, tank blanketing concern	Temperature high alarm TAH-2107	Include E-2107 bypass valve in the passing-valve survey scope	K. Rao	2026-12-31	Not Started
H7-06	Reverse Flow	Non-return valve NRV-2104B fails to seat	Reverse rotation of standby pump, coupling damage	NRV-2104A/B, discharge isolation procedure in SOP-CDU2-014	Add NRV-2104A/B to the 4 yearly strip-and-inspect register	V. Nayak	2027-03-31	Not Started

NOTES ON OVERDUE ACTION H7-03

The relief load recalculation for the blocked discharge scenario on the P-2104 discharge header indicates that the required relief capacity exceeds the installed capacity of PSV-2141 by an estimated 8%. The valve was pop tested and recertified under WO-26-13001 on 2026-06-15 and passed at its set pressure of 26.4 kg/cm2g, but a pop test confirms set pressure only and does not address capacity. Re-rating requires either an orifice change or a second valve, and the tie-in would fall within the zone restricted by SC-2026-11.

Because this action is now more than 60 days past its due date of 2026-06-30, it must be escalated to the Refinery Process Safety Committee with a revised target date and an interim risk statement.